

REMOTE VALVE CONTROL MECHANISM

IoT-Based Smart Irrigation System for Banana Farms

Area: 3 Acres (Banana Farm)

1. Project Overview

This project proposes a low-power, cost-effective remote valve control system for irrigation management in a 3-acre banana farm. The system uses LoRa-based wireless communication to enable a farmer to open or close water valves remotely without any manual effort at each valve location.

The farm layout is approximately 300X50mts A single Master Node is placed centrally and communicates with multiple Slave Nodes distributed across the field.

2. Project Objectives

- Control irrigation valves remotely (Open / Close) via mobile commands
- Design an ultra-low-power system to minimise battery and solar panel costs
- Ensure system reliability for at least 3 years of continuous operation
- Keep the solution budget-friendly using off-the-shelf components
- Use LoRa long-range wireless protocol for reliable field communication

3. System Architecture

The system follows a Master-Slave architecture over LoRa:

Component	Role	Location
Master Node	Receives SMS/GSM commands from farmer, relays to slave nodes via LoRa	Central field position
Slave Nodes (x4+)	Receives valve commands, drives motor actuator, reports status	Distributed across 3 acres

Component	Role	Location
GSM Module	Enables farmer to send ON/OFF commands via mobile SMS	Master Node
LoRa Module	Long-range wireless link between master and slaves	All nodes
DC Gear Motor	Physically opens/closes the valve	Each slave node
Limit Switches	Detect fully open / fully closed valve position	Each slave node

4. Slave Node Design

4.1 Components

1. Microcontroller (MC): ESP32 / NodeMCU
2. LoRa Module: SX1276 (or similar) operating in Long Range Mode
3. Motor Driver: BTS7960 H-Bridge
4. DC Gear Motor: 12V, 10-30 RPM, 20-25 kg.cm torque
5. Limit Switches (x2): For detecting valve open/close status
6. Power: 12V Battery + 6W Solar Panel + Charging Controller + Buck Converter (12V to 5V)
7. MOSFET switches: To cut power to peripherals during sleep

4.2 Slave Node Bill of Materials (BOM)

Component	Specification	Estimated Cost (INR)
Microcontroller (ESP32)	NodeMCU / ESP32	150 – 200
LoRa Module + Antenna	SX1276 or equivalent	200 – 250
Motor Driver	BTS7960	200
DC Gear Motor	12V, 10-30 RPM, 20-25 kg.cm	400
Limit Switches	2 x ₹20	40
12V Battery	Amp-hours based on power requirement	Under ₹500
6W Solar Panel + Charge Controller	Along with charge controller	Under ₹500
Buck Converter (12V to 5V)	For logic circuit supply	Under ₹100
MOSFET	For power switching	10
Misc (PCB, connectors, casing)	—	~100 – 200

Estimated Slave Node Total: Under ₹1,500 per node

5. Master Node Design

5.1 Components

8. Microcontroller (MC): ESP32 / NodeMCU
9. LoRa Module: Long Range Mode
10. GSM Module: For mobile communication (farmer sends SMS commands)
11. Power: Solar Panel + Battery (based on power requirement)

5.2 Master Node Bill of Materials (BOM)

Component	Specification	Estimated Cost (INR)
Microcontroller (ESP32)	NodeMCU / ESP32	150 – 200
LoRa Module	Long Range, with antenna	200 – 250
GSM Module	SIM800L or equivalent	500 – 700
Solar Panel + Battery	Based on power requirement	As required

6. Power Management Strategy

A key design goal is ultra-low power consumption to maximise battery life and minimise solar panel size. The following strategies are employed:

6.1 Deep Sleep Architecture

- Controller uses deep sleep mode between communication windows
- RTC (Real-Time Clock) is used to schedule wake-up intervals
- During sleep, the RTC controls when the node wakes and when farmers' messages can be relayed
- All peripheral modules (LoRa, motor driver, GSM) are powered via MOSFET switches — power is cut when not in use

6.2 Continuous Power Requirements

Continuous power is ONLY required for:

- Microcontroller (during active operation)
- LoRa module (during transmit/receive windows)
- Motor driver (only when actuating the valve)

6.3 Actuator Power Profile

The DC Gear Motor only draws power during the valve actuation cycle (approx. 10-20 seconds per open/close operation). As a gear motor holds its position mechanically, it does not consume continuous power to maintain valve position. This significantly reduces average power draw.

6.4 Solar & Battery Sizing

- Solar panel: 6W (sufficient for low-duty-cycle operation)
- Battery: 12V, capacity sized based on calculated power requirement
- Buck Converter: Steps 12V down to 5V for logic circuits
- Solar cost can be further reduced by minimising all standby power consumption

7. Communication Flow

Step	Description
1	Farmer sends SMS command (e.g., 'VALVE1 ON' or 'VALVE2 OFF') to Master Node GSM number
2	Master Node GSM module receives the SMS and parses the command
3	Master Node wakes up LoRa module and transmits command to the target Slave Node
4	Slave Node receives LoRa packet, wakes up motor driver, and actuates the valve
5	Limit switch confirms valve reached open or closed position
6	Slave Node sends status acknowledgement back to Master via LoRa
7	Master sends confirmation SMS back to farmer (optional)

8. Farm Field Layout

The 3-acre banana farm is approximately 120m (N-S) x 50m (E-W). Node placement:

Node	Type	Position	Coverage
Node 0 (Master)	Master	Centre of field (~50m from south, ~60m from north)	Full field via LoRa
Node 1 (Slave)	Slave	North area (~120m from south end)	North irrigation zone
Node 2 (Slave)	Slave	North-centre (~80m from south end)	North-centre zone

Node	Type	Position	Coverage
Node 3 (Slave)	Slave	South-centre (~30m from south end)	South-centre zone
Node 4+5 (Slave)	Slave (pair)	South end, side by side	South irrigation zones

9. Challenges & Mitigation

Challenge	Mitigation Strategy
Power requirement = cost	Ultra-low-power design; deep sleep; MOSFET-gated peripherals; small 6W solar panel
Long battery life requirement	Gear motor doesn't hold continuous power; RTC scheduling; all modules powered off when idle
Reliable long-range communication	LoRa operates in long-range mode; suitable for open farm fields
Solar panel cost reduction	Achieved by minimising power requirements across all nodes
Valve position accuracy	Dual limit switches on each actuator confirm fully open / fully closed state

10. Technology Stack

Layer	Technology / Component
Microcontroller	ESP32 / NodeMCU
Wireless Protocol	LoRa (Long Range, Sub-GHz ISM band)
Mobile Communication	GSM (SIM800L or similar) — SMS based
Motor Control	BTS7960 H-Bridge Driver
Actuator	12V DC Gear Motor (10-30 RPM, 20-25 kg.cm)
Power Management	12V Battery + 6W Solar + Charge Controller + Buck Converter + MOSFET switching
Position Sensing	Mechanical Limit Switches (x2 per node)
Sleep/Wake	RTC-based scheduling + ESP32 deep sleep

11. Proposed Project Timeline

Phase	Activity	Duration
Phase 1	Component procurement and hardware testing	Week 1-2
Phase 2	Slave node firmware development (LoRa + Motor control)	Week 3-4
Phase 3	Master node firmware (GSM + LoRa gateway)	Week 5-6
Phase 4	Power management & deep sleep integration	Week 7
Phase 5	Field testing and calibration on farm site	Week 8-9
Phase 6	Documentation, review, and final demonstration	Week 10

12. Summary

The Remote Valve Control Mechanism is a practical, low-cost IoT solution designed specifically for small to medium-scale agriculture. By combining ESP32 microcontrollers, LoRa long-range wireless, GSM mobile connectivity, and intelligent power management, this system enables a farmer to control all irrigation valves across a 3-acre banana farm from their mobile phone — with a target operational life of 3+ years on solar power.

The design prioritises:

- Simplicity — farmer just sends ON/OFF SMS
- Low power — deep sleep + MOSFET-gated modules + gear motor hold
- Low cost — estimated under ₹1,500 per slave node
- Reliability — limit switches confirm valve status; LoRa provides robust wireless link



